FSoD-Net: Full-Scale Object Detection From Optical Remote Sensing Imagery

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***Abstract*— Object detection is an essential task in computer vision. Recently, several convolution neural network (CNN)-based detectors have achieved a great success in natural scenes. However, for optical remote sensing images with a large scale of view, lower proportion of foreground target pixels and drastic differences in object scale present considerable challenges. To address these problems, we propose a novel one-stage detector called the full-scale object detection network (FSoD-Net) which consists of proposed multiscale enhancement network (MSE-Net) backbone cascaded with scale-invariant regression layers (SIRLs). First, MSE-Net provides the multiscale description enhancement by integrated the Laplace kernel with fewer parallel multiscale convolution layers. Second, SIRLs contain three different isolated regression branch layers (i.e., corresponding to small, medium, and large scales), which make default discrete scale bounding boxes (bboxes) cover full-scale object information in regression procedure. A novel specific scale joint loss is also designed that uses the softmax function combined with a strong L_1-norm constraint in each regression branch layer. It can further speed up the convergence and improve the classification scores of predicted bboxes. Finally, extensive experiments are carried on challenge data sets of large-scale dataset for object detection in aerial images (DOTA) and object detection in optical remote sensing images (DIOR) which contain multiple instances from different imaging platforms, and these results demonstrate that FSoD-Net can achieve better performance than other state-of-the-art one-stage detectors, and it can reach a mean average precision (mAP) of 75.33% on DOTA and 71.80% mAP on DIOR, respectively. Especially, the average precision (AP) of tiny object detection can improve 10%–20% approximately.**

***Index Terms*— Convolution neural network (CNN), full-scale object detection, one-stage detector, optical remote sensing.**

I. Introduction

OBJECT detection is a fundamental task in computer vision, and it can be used for intelligent surveillance system such as military reconnaissance, security monitoring, and traffic controlling [1]–[6]. To avoid the effects of scale and category variance of multiple objects and unpredictable background influences in object detection task, the traditional methods usually use the several scale-invariant feature descriptors including the local binary pattern (LBP) [7], scale-invariant feature transform (SIFT) [8], histogram of orientation gradient (HOG) [9] and so on, cascaded with a specific classifier to achieve the object detection. However, since these feature descriptors have an insufficient feature generalization ability, traditional methods [7]–[15] often appear missed detection and massive false alarms. Compared with the traditional detection methods, benefiting from the powerful feature extraction ability of convolution neural network (CNN), several CNN-based detectors, the single shot multibox detector (SSD) [16], You Only Look Once (YOLO) [17], RetinaNet [18], receptive field block (RFB)-Net [19], Faster R-CNN [20], feature pyramid network (FPN) [21], spatial pyramid pooling (SPP)-Net [22], and so on [23]–[30], have achieved a great success in processing nature scene images. Recently, a lot of researchers consider applying these mentioned famous one- or two-stage detectors in optical remote sensing object detection. However, there are different detection scenarios between optical remote sensing and nature scenes as illustrated in Fig. 1. First, in Fig. 1(a) and (b), because of the large-view-scale overlooking observation, the optical remote sensing scenes would contain more unseen interferences from complex background than nature scenes, which can make more false alarms appear. Second, in Fig. 1(c), the corresponding foreground–background ratios of six public object detection data sets are counted to exhibit differences between optical remote sensing and nature scene object detection tasks. Here, the foreground–background ratio represents the proportion of the interested object area divided by the whole image area, and for fairly comparing, optical remote sensing and natural scene images are regulated into 800×800 pixels. The lengths of color bars in Fig. 1(c) indicate the scopes of object scale variance in the different data sets, and it is obvious that the public optical remote sensing data sets large-scale dataset for object detection in aerial images (DOTA) [47] and object detection in optical remote sensing images (DIOR) [48] own the larger scale span than that in nature scene data sets. Consequently, the default bounding boxes' (bboxes) setting for continuous object scale description within a large-scale span and the effective design of regression layers are important elements to influence the optical remote sensing object detection performance. Then, in Fig. 1(c), the different colors in each bar mean the object scale ranges and the length of each color block represents the proportion of object quantity in each scale range. For instance, the pink and light-yellow blocks represent the tiny objects with 0–50 square rooted pixels. Combined with Fig. 1(a) and (c),

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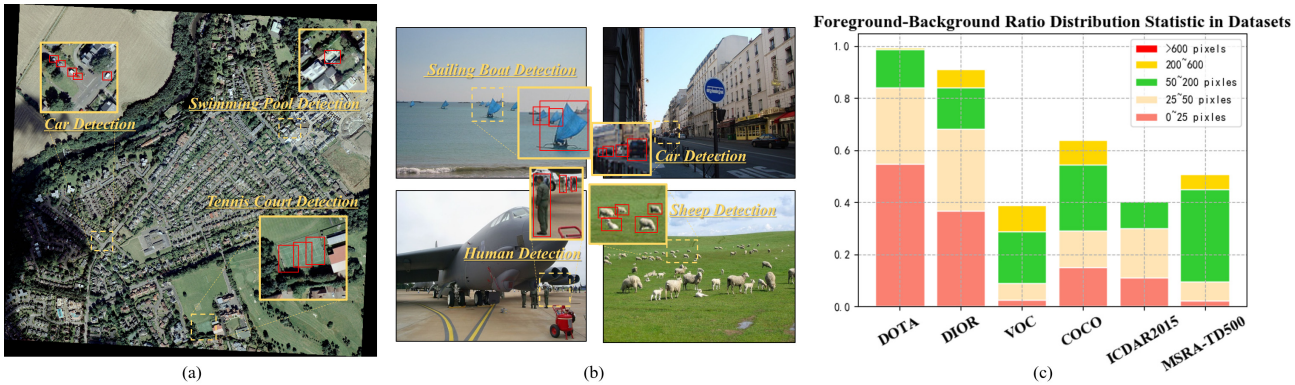
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(a) (b) (c)

Fig. 1. Object detection in optical remote sensing and nature scenes. (a) Optical remote sensing object detection. (b) Nature scene object detection. (c) Object scale distribution in optical remote sensing data sets of the DOTA and DIOR and nature scene data sets of PASCAL Visual Object Classes (VOC), Microsoft Common Objects in Context (COCO), International Conference on Document Analysis and Recognition 2015 (ICDAR2015), and MSRA-text detection 500 (TD500) (i.e., all the foreground-background ratios are calculated based on regulated 800 × 800 pixels image).

we can see that tiny object detection frequently occurs in the large-view-scale optical remote sensing scenes. It often needs to extract approximately a dozen pixels from the large width overlooking scenes with hundreds of megapixels, which can cover areas around >32 km^2.

Facing problems as mentioned before, the famous CNN-based one- or two-stage detectors cannot be directly used for optical remote sensing object detection, because they still have the room to further explore tiny object detection, false alarm removing, and continuous object scale representation within the large-scale span. Thus, in recent years, a lot of works [31]–[46] are concentrated on the CNN-based large-view-scale optical remote sensing object detection. Using transfer learning and data augmentation, Han *et al.* [33] proposed a more efficient and robustness geospatial object detection framework based on a two-stage Faster R-CNN framework with a visual geometry group (VGG) backbone. Subsequently, Li *et al.* [34] proposed a multiscale deep feature embedding structure based on Faster R-CNN to adapt the variations in scale and facilitate both large and small ship detection. Furthermore, with the aim of balancing the detection accuracy and efficiency, some one-stage detection methods are often applied. For example, Zhuang [36] *et al.* used SSD detection framework combined with a multiscale feature fusion strategy and a Darknet-53 backbone to achieve object detection, with a particular focus on small objects. Then, to realize multiscale object detection from complex optical remote sensing images, Tayara *et al.* [37] also proposed a one-stage detector for a set of discrete bboxes' regression and prediction based on the densely connected feature pyramid network. In previous works, the powerful CNN backbone is often used for one- or two-stage detectors to facilitate multiple categories' objects' classification and avoid false alarms from complex optical remote sensing scenes. However, it is hard to trade-off multiscale object localization ability with feature extraction ability throughout the depth of the whole network. For instance, despite deep feature layer owning a powerful classification ability, the continuous convolution and pooling operations will generate an excessive context information and weaken their localization information. As a result, missed detection of tiny objects often appears in deep feature layers. On the other hand, although the high resolution of shallow layers can facilitate tiny object detection, they would produce a lot of false alarms because of the feature description limitation. Therefore, how to set up powerful object detection backbone and construct effective regression layers becomes challenging for large-scale span of optical remote sensing object detection.

In this article, for achieving fast and accurate optical remote sensing object detection by the CNN-based detector, a novel one-stage full-scale object detection network (FSoD-Net) is proposed. It consists of a powerful multiscale enhancement network (MSE-Net) backbone cascaded with scale-invariant regression layers (SIRLs) as shown in Fig. 2. First, a novel lightweight MSE-Net is proposed to achieve a powerful feature extraction and avoid submergence of tiny object information in backbone by Laplace spatial enhancement (SE) cascaded with fewer convolution layers. Followed by the proposed MSE-Net backbone are the newly designed SIRLs. They consist of three isolated regression branch layers corresponding to different scales (i.e., small, medium, and large), and the purpose of their structure is to achieve the large-scale span object description by predefined discrete bboxes' regression procedure. Furthermore, related to newly designed SIRLs, a specific scale joint loss is proposed to further speed up the convergence and improve the classification scores of the predicted bboxes. Finally, the challenging DOTA [47] and DIOR [48] data sets are used to demonstrate that the proposed FSoD-Net can address these problems as discussed above. The extensive experiments prove that the proposed FSoD-Net outperforms other state-of-the-art CNN-based one-stage detectors, and it can achieve a mean average precision (mAP) of 75.33% on DOTA [47] and 71.80% mAP on DIOR [48], respectively. Furthermore, related to tiny object detection, FSoD-Net can achieve the 10%–20% average precision (AP) improvement compared with the state-of-the-art methods. In general, the contributions of this article can be summarized as follows:

1) Inspired by the object detection network SSD, a novel one-stage detection network called FSoD-Net is proposed for multiscale object detection in complex optical remote sensing scenes. It can achieve the rapid and accurate object detection with less false alarms, and the comparison results demonstrate that our pro-

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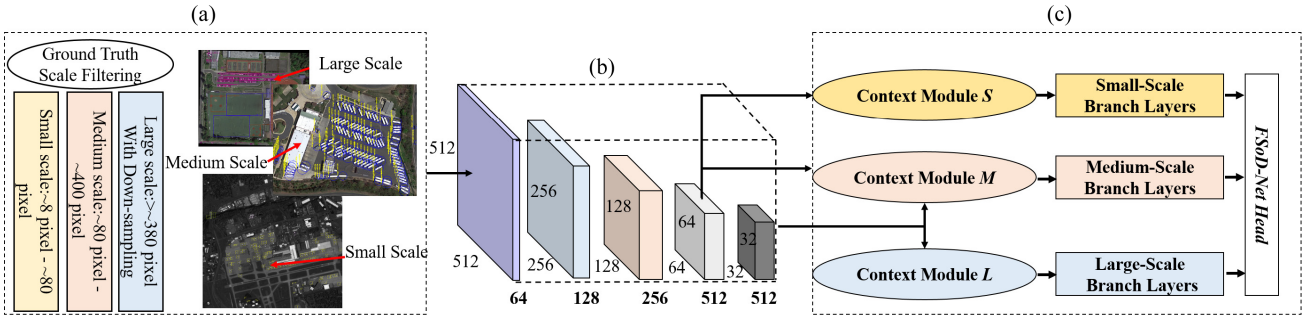


Fig. 2. Framework of the proposed FSoD-Net. (a) Scale division of the training data. (b) Simple schematic of the MSE-Net backbone network. (c) Schematic of the proposed SIRLs.

posed FSoD-Net is superior to the other state-of-the-art CNN-based one-stage object detectors.

2) With the aim of improving basic feature extraction ability while avoiding false alarms, a powerful lightweight backbone MSE-Net is proposed. Especially for tiny object detection, MSE-Net backbone can prominently enhance the information extracted on weak or tiny object from the input optical remote sensing images, and it can also provide the effective feature description for any scale object detection.
3) To deal with the large-scale span object detection from complex optical remote sensing scenes, a novel SIRLs and their corresponding specific scale joint loss are proposed. They can make predefined discrete bboxes cover the full-scale object information in regression procedure and further speed up the convergence and improve the classification scores of the predicted bboxes by L_1-norm constraint.

II. Related Work

Owing to the requirement of real-time object detection from unmanned embedded platforms (i.e., unmanned aerial vehicles (UAVs) or satellites), the object detection algorithm has to take into account the high accuracy and efficiency for practical applications automatically (i.e., ship search and rescue, intelligent traffic controlling, and so on). Although the one-stage CNN-based detectors have a certain real-time object detection ability compared with two-stage detectors, they still provide a lower detection accuracy which cannot be used for practical system. Therefore, aiming to build a powerful one-stage object detector for optical remote sensing object detection, many efforts have been made and they are all concentrated on two parts: 1) to construct a lightweight backbone and 2) set up suitable regression layers. Then, in view of these two parts, we would elaborate the related works of this article.

A. Lightweight Backbone for One-Stage Detectors

Initially, CNN architectures (i.e., AlexNet [49], GoogleNet [50], and LeNet-5 [51]) are widely used for classification tasks. Then, with the development of CNN classification technology, a lot of networks go into deep layers to obtain more powerful classification ability such as VGG-16 [52], VGG-19 [52], ResNet-50 [53], DarkNet-53 [28], and ResNet-101 [53]. These powerful deep CNNs are usually used as backbone for object detection to be against with the complex background interferences. Although the powerful deep CNN architecture can achieve good object classification and avoid false alarm production, it is not friendly to tiny object detection, which is mentioned and analyzed in Section I. Therefore, Qin *et al.* [58] used the Yolo-v3 framework and considered to avoid the tiny objects' information being weakened in the deep layers. They set up a backbone by Darknet-9 cascaded with receptive field enhancement module, which can construct the context information to enhance the tiny object feature description by different rates of atrous convolutions. Then, Pang *et al.* [59] set up the fast tiny object detection backbone by referring ResNet [53] framework and integrated global attention module at the end of Tiny-Net backbone to fuse different pyramid scale features. Moreover, Li *et al.* [54] proposed a more effective backbone network of DetNet for object detection, which can obtain deep feature maps with high spatial resolution for tiny object detection and keep a powerful classification ability by designed feature pyramid bottleneck structure. In general, since tiny object detection usually occurred in optical remote sensing object detection task, a lot of works are not blindly increasing the depth of backbone network, and they almost focus on enhancing the shallow layer feature description ability by context information for tiny object detection. Then, in this article, we pay more attention to increasing the width of the backbone network instead of blindly pursuing the depth, so as to improve the feature description ability in arranged fewer convolution layers. Besides, we also consider enhancing the tiny object information in the backbone network to alleviate their information weakened phenomenon.

B. Regression Layers Designing for One-Stage Detectors

With respect to the optical remote sensing object detection, the performance is severely restricted by large-scale span variance of detected objects. Facing multiscale object detection in the CNN-based one-stage detector, a lot of research works [60]–[65] are using FPN [21] structure to generate specific multiscale regression layers for default multiscale bboxes' regression and prediction. For example, Zhang *et al.* [63] proposed the attention-guided network based on the Yolo-v3

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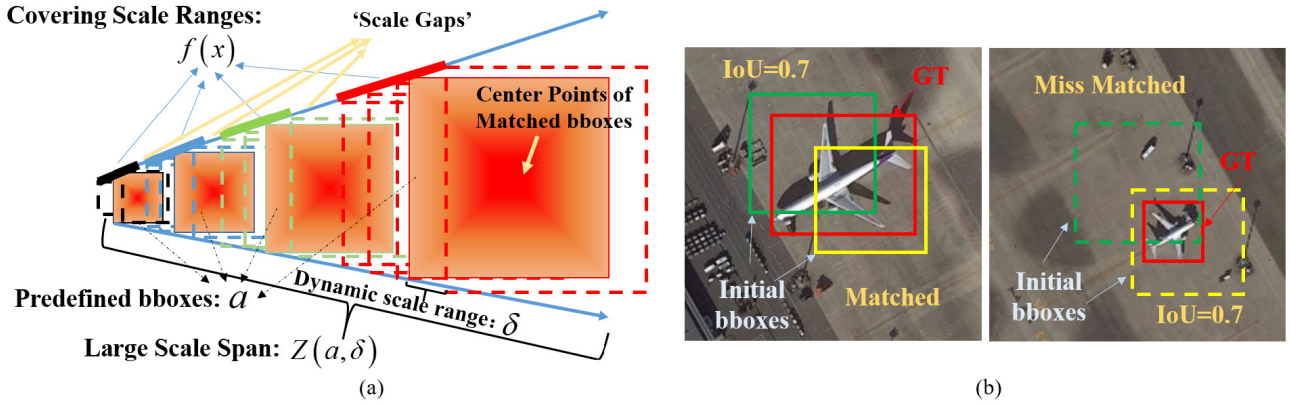


Fig. 3. Design ideal and principle illustration of the proposed FSoD-Net. (a) Problem of predefined multiscale discrete bboxes for continuous scale representation within a large-scale span. (b) Principle of the IoU threshold constraint in the training phase.

framework, and they set up multiscale regression layers by adjacent scale layer fusion of feature concatenation and channel attention. Next, Etten [64] proposed a one-stage optical remote sensing object detector called You Only Look Twice (YOLT), which designed multiscale predicted layers for large or tiny object detection. Moreover, Wang *et al.* [65] referred the SSD multiscale predicted layers and used feature fusion way of top–down connection between layers achieving the multiscale object detection. In summary, although abundant of new regression and predicted layers are designed for multiscale optical remote sensing object detection, they often provide bad detection performances, because a specific FPN structure has a limitation to cover full-scale range of optical remote sensing object by default discrete scale bboxes. Thus, we rethink the large-scale span optical remote sensing object representation and refer to previous works of enhancing the multiscale feature description ability to construct suitable and effective regression layers.

III. Proposed Methodology

In this section, we describe FSoD-Net in detail. First, Section III-A introduces the overview of the proposed FSoD-Net. Then, Section III-B presents the MSE-Net backbone. Next, Section III-C describes the structure of the SIRLs. Finally, the specific scale joint loss of FSoD-Net is introduced in Section III-D.

A. Overview of Proposed FSoD-Net

Before discussing the particular architecture of FSoD-Net as shown in Fig. 2, we should clarify the main design ideal first. With regard to optical remote sensing object detection within large-scale span, a vast scale change is illustrated in Fig. 1(c), which is the main reason to restrict the CNN-based one-stage detectors' performance. Recently, several research works [66]–[69] have also pointed out this problem. They try to fit the large-scale span of optical remote sensing object by densely predefined discrete multiscale bboxes corresponding to the specific FPN structure. However, the detection results are still unsatisfactory.

In Fig. 3(a), the blue axes are the large-scale span $Z(a, \delta)$. Here, a represents a set of predefined discrete multiscale bboxes which are shown in multiscale orange rectangles of Fig. 3(a), and δ means the dynamic update scale range of the predefined discrete multiscale bboxes. Then, δ variance is driven by the gradient update direction, and it depends on matched scales of training data by IoU threshold constraint as shown in Fig. 3(b) and the red areas of multiscale orange rectangles in Fig. 3(a). Therefore, the large-scale span $Z(a, \delta)$ in Fig. 3(a) can be expressed as

$$Z(a,\delta) = \{f(x) | a_{\min} - \delta < f(x) < a_{\max} + \delta\} \quad \text{s.t. IoU} > T. \tag{1}$$

In (1), $f(x)$ represents scale ranges of the predefined discrete multiscale bboxes after regression procedure as shown in different color lines of Fig. 3(a). If these discrete bboxes are defined in a narrow-scale range or a few scale layers are used for regression, the rest of the scale ranges in a large-scale span would be ignored in the model training phase. Obviously, it is hard to represent objects covering a large scale span by predefined discrete multiscale bboxes based on a few scale layers of the FPN structure. Therefore, from Fig. 3(a), we consider to divide the range of large-scale span into several fine-scale segments, which can be expressed as (2)

$$\begin{aligned} &Z(a,\delta) = Z_1(a_1,\delta_1) \cup Z_2(a_2,\delta_2) \cup \cdots \cup Z_n(a_n,\delta_n) \\ &\text{s.t. } \forall Z_n(a_n,\delta_n) \cap Z_{n-1}(a_{n-1},\delta_{n-1}) \neq \emptyset, \ 1 < n \leq K \end{aligned} \tag{2}$$

where in (2), each fine-scale range of $Z_n(a_n, \delta_n)$, $n = 1.2, \ldots, K$ can be easily described and fit by the predefined discrete multiscale bboxes corresponding to several regression layers. Thus, based on this theory, SIRLs in FSoD-Net as shown in Fig. 2(c) are composed of the small-, medium-, and large-scale branch layers, and in each branch layer, it also contains several newly designed regression layers to avoid the ground truth from under- or over-fitting within local scale distribution. Then, related to the setup of SIRLs' structure, the data preprocessing used k-means clustering algorithm [17] to generate three segments: small-scale labeled object (i.e., square root areas (sqrt areas) of ∼8 to ∼80 pixels in data set); medium-scale labeled objects (i.e., the sqrt areas of ∼80 to ∼400 pixels in data set), and large-scale labeled objects (i.e., the sqrt areas of more than 400 pixels in data set), and each of which has a smaller scale range

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variation than the full-scale data set does. When the scale filtering preprocessing is completed, we would refer these generated scale ranges to carefully define initial bboxes. In addition, as shown in Fig. 3(a), we also need to consider the "scale gap" between inter- and intra-adjacent scale of small, medium, and large scales, which can lead to relatively insufficient samples for model training under the strict IoU threshold setting. Here, the scale compensation strategy can be expressed as

$$m(Z_n(a_n, \delta_n) \cap Z_{n-1}(a_{n-1}, \delta_{n-1})) = \frac{1}{2} m(Z_n(a_n, \delta_n)). \quad (3)$$

In (3), $m(\cdot)$ denotes a measure-based operation, and the scale compensation strategy of (3) means each fine-scale range has a roughly 50% overlap with its adjacent scale ranges, whether in inter- or intra-scales. Finally, after scale filtering preprocessing and careful scale compensation predefined bboxes' setting, each segment of different scale labeled samples would be passed through the powerful MSE-Net and then fed into the regression layers of the corresponding scale via IoU selection. In Section III-B, the MSE-Net backbone, as shown in Fig. 2(b), is described in detail. The SIRLs, as shown in Fig. 2(c), is elaborated in Section III-C. In Section III-D. the specific scale joint loss will be described in detail.

B. MSE-Net Backbone

The idea of focusing on increasing the width of backbone instead of blindly pursuing the depth was followed; meanwhile, enriching the weak or tiny object information, a powerful lightweight MSE-Net backbone is proposed. Here, MSE-Net is generated by two element structures: 1) SE module and 2) multiscale convolution (MSC) layer as shown in Fig. 4(a) and (b). Then, the detail structure of MSE-Net is also illustrated in Table I.

First, as illustrated in Fig. 4(a), before the input image is fed into the MSC layers for multiscale feature extraction, an SE module in parallel with a set of sequential convolutions is applied to enhance weak or tiny objects and avoid them getting lost in following the MSC layers, where a trainable 3×3 Laplace kernel is embedded into the SE module. When the Laplace kernel is applied to a three-channel RGB input image, the weak or tiny objects presented in each channel are sharpened by means of Laplace feature enhancement. Next, these enhanced features of three input channels are fed into a 3×3 convolution to increase the number of channels from 3 to 64. At the same time, the original input image is also fed into a 3×3 convolution followed by a 3×3 dilated convolution to extract the fine-scale context information to support weak or tiny object feature description. Here, these sequential convolutions applied to the original input image can produce feature maps with 64 channels. After a batch normalization (BN) operation, these 64-channel feature maps from both the branches are added together. In general, the SE module including a trainable Laplace kernel with parallel sequential convolutions can improve the feature representation ability for shallow layers, which is vital for tiny object detection.

Second, in Fig. 4(a), after a rectified linear unit (ReLU) operation, the enhanced feature maps are fed into the MSC

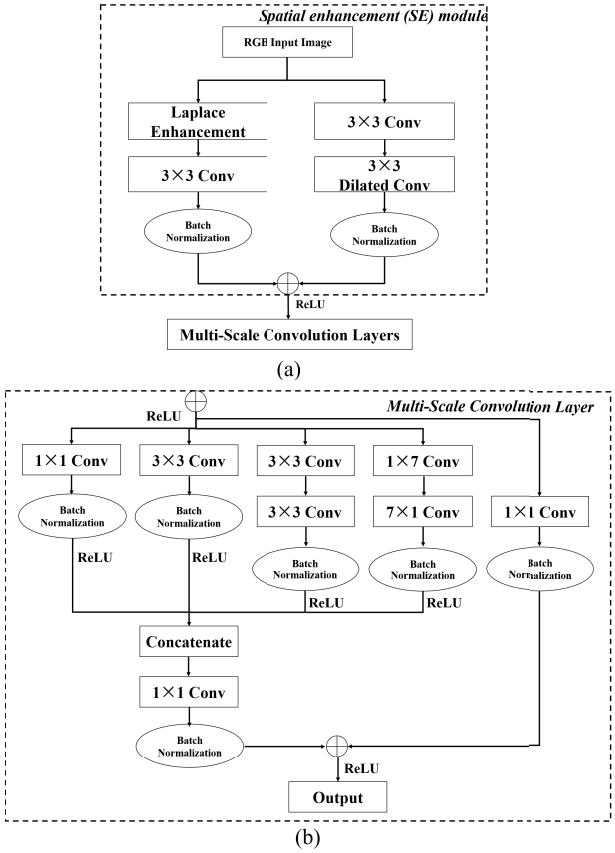


Fig. 4. Structure of the proposed MSE-Net. (a) Spatial SE module. (b) Structure of the MSC layer.

layers, as shown in Fig. 4(b). To improve the multiscale feature extraction capability of the network, the MSC layer uses parallel convolution kernels with different sizes to generate feature maps with different receptive fields. Specifically, kernel sizes of 1×1, 3×3, 5×5, and 7×7 are used in one MSC layer for parallel convolution operations. Here, the 5×5 convolution is achieved by two sequential 3×3 convolutions, and the 7×7 convolution is achieved by means of a 1×7 convolution and a 7×1 convolution, which can reduce the parameters of the MSC layers. Once the convolution operations are completed, these BN and ReLU operations are applied to the parallel convolutional feature maps before they are concatenated. Next, a 1×1 convolution is applied to the concatenated feature maps to achieve channel compression and cross-channel information interactions. Then, before the final ReLU operation is applied to produce the output, a skip connection operation is used to prevent the gradient anomalies and achieve the multiscale feature fusion. At the end, the MSE-Net backbone contains five MSC layers in total, as shown in Table I, through which the basic feature description MSE-Net not only can facilitate the weak or tiny object detection but also be efficient with any scale object feature extraction.

C. Scale-Invariant Regression Layers

Referring to multiscale regression and prediction layers in SSD [16], as well as the design idea mentioned before, a newly designed SIRLs' structure is proposed. The SIRLs' structure

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TABLE I
DETAILED ARCHITECTURE AND PARAMETERS OF THE MSE-NET BACKBONE

Layer Name	Architecture	Input Size	Output Size	Parameters
SE-Module	3×3 Laplace Kernel	(512,512,3)	(512,512,3)	(3×3×3)×3
	3×3 Conv	(512,512,3)	(512,512,64)	(3×3×3)×64
Sequential Conv	3×3 Conv	(512,512,3)	(512,512,64)	(3×3×3)×64
	3×3 Conv, Rate=2	(512,512,64)	(512,512,64)	(3×3×64)×64
MSC Layer 1		(512,512,64)	(512,512,64)	
Pooling Layer 1	3×3 Conv, Stride=2	(512,512,64)	(256,256,128)	(3×3×64)×128
MSC Layer 2		(256,256,128)	(256,256,128)	
Pooling Layer 2	3×3 Conv, Stride=2	(256,256,128)	(128,128,256)	(3×3×128)×256
MSC Layer 3		(128,128,256)	(128,128,256)	
Pooling Layer 3	3×3 Conv, Stride=2	(128,128,256)	(64,64,512)	(3×3×256)×512
MSC Layer 4		(64,64,512)	(64,64,512)	
Pooling Layer 4	3×3 Conv, Stride=2	(64,64,512)	(32,32,512)	(3×3×512)×512
MSC Layer 5		(32,32,512)	(32,32,512)	

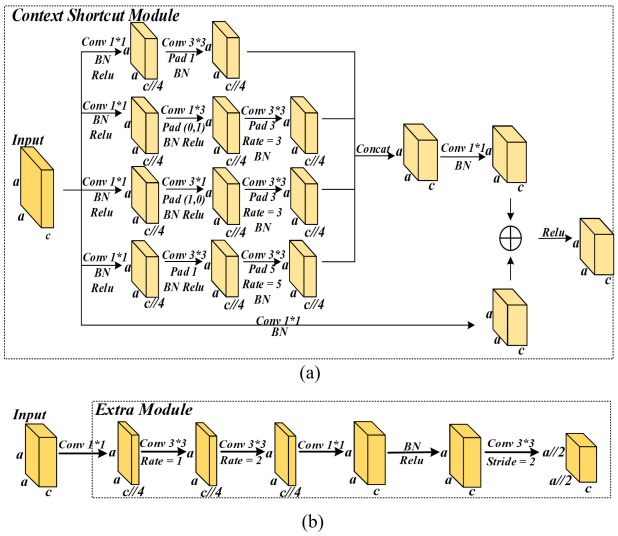


Fig. 5. Newly designed context shortcut module and extra module. (a) Context shortcut module. (b) Extra module.

includes three isolated regression branch layers, which can be used to fit the multiscale objects by the predefined discrete multiscale bboxes' regression based on the scale compensation strategy within the large-scale span. Here, in each of small-, medium-, or large-scale branch, the discrete predefined bboxes' setting approach is similar to SSD [16], and these bboxes are defined in terms of *sizes* and *aspect ratios*. Then, in FSoD-Net, the IoU threshold is set to 0.5 for model training. Since scale variances in small-, medium-, and large-scale branches are different, each scale branch layer has a different structure to fit their local fine-scale distributions. Therefore, related to SIRLs' structure, we would describe each scale branch layer individually. Before elaborating the detail structure of each scale branch, we would introduce the new context shortcut and extra modules first, because three different scale branch layers are all constructed by these two modules.

As shown in Fig. 5(a), the context shortcut module consists of a group convolution consisting of four parallel convolutions based on bottleneck structure. Here, with regard to the input feature maps of the designed context shortcut module, there are several parallel 1×1 convolutions, batch normal, and Relu operations applied, which can refine the input feature maps for bboxes' regression. Next, followed by 1×1 convolutions, the diverse 1×3, 3×1, and 3×3 convolutions with different dilated rates of atrous convolutions are used to further improve the multiscale feature description ability. Finally, 1×1 convolution is applied for concatenated multiscale feature maps with four groups, and then it is added back into the input feature maps, which have been applied the 1×1 convolution and batch normal, to generate the output feature layers by Relu. Then, as shown in Fig. 5(b), the designed extra module is composed of several cascaded convolutions. First, the 1×1 convolution is applied for input to compress the feature channels. Next, followed by the channel compression, cascaded 3×3 convolutions with different dilated rates and another 1×1 convolution are applied for recovering the channel number. At the end of the extra module, the 3×3 convolution with stride $= 2$ is used to achieve feature map down-sampling and capture a larger receptive field for the suitable predefined scale bboxes' regression. In general, the context shortcut module is designed for further improving the multiscale feature description ability of each regression layer, and the extra module is designed for down-sampling feature maps to enlarge the receptive field for larger scale object feature description. Next, the three different scale branch layers are introduced below:

S-branch: The S-branch is designed to detect objects with sqrt areas from around 8–80 pixels. They are considered to be small objects in optical remote sensing images. The small-scale object has smaller scale variance than medium and large scales. Therefore, the structure of the S-branch consists of five regression layers with the same size as shown in the left side of Figs. 6(a) and 7. In Fig. 6(a), diversely colored lines represent the corresponding fine-scale ranges of the default bboxes, and these pink rectangles represent the overlapping of scale compensation strategy in the S-branch. In Fig. 7, MSC layer 4 in the backbone is used to construct the S-branch. First, MSC layer 4 is fed into a context shortcut module, and then, the different rates of 3×3 convolutions are used to generate five regression layers for small-scale object localization and classification regression.

M-branch: The M-branch is designed to detect objects with sqrt areas from about 80–400 pixels. They are considered to be medium-scale objects in optical remote sensing images. The medium-scale objects show more drastic scale variation

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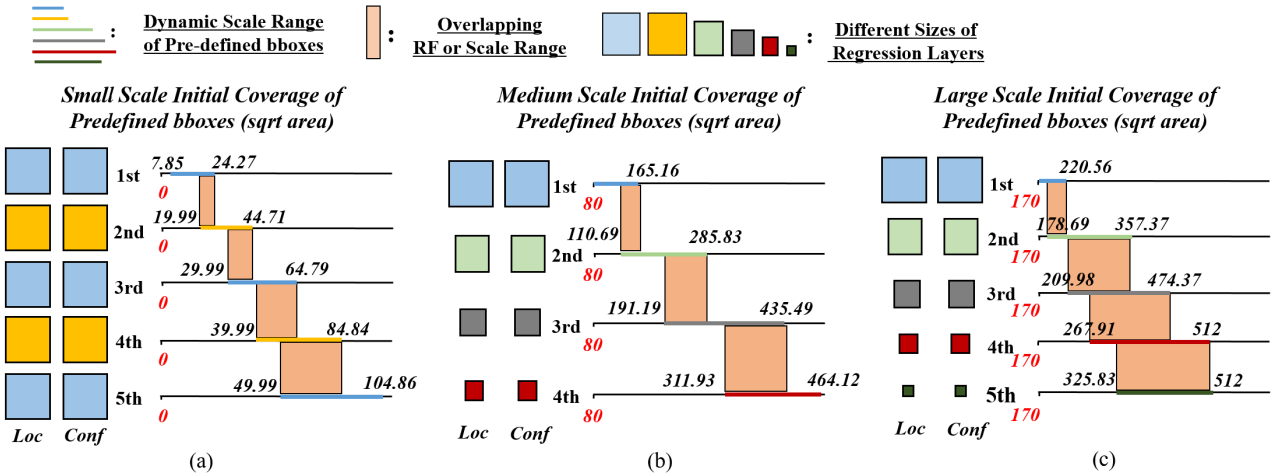


Fig. 6. Scale compensation strategy in the SIRLs. (a) Small scale, (b) medium scale, and (c) large scale initial coverage of predefined bboxes (sqrt area).

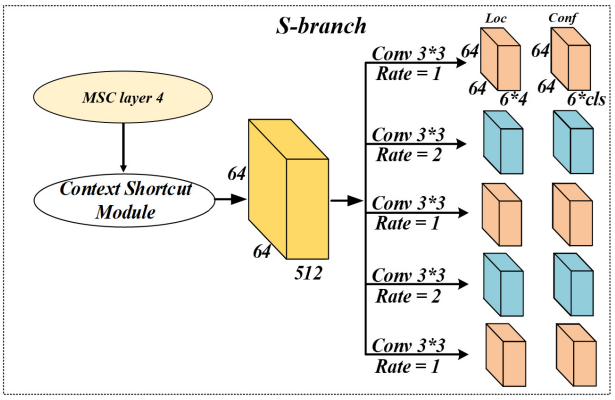


Fig. 7. Detailed structure of the small-scale object regression branch layers in SIRLs.

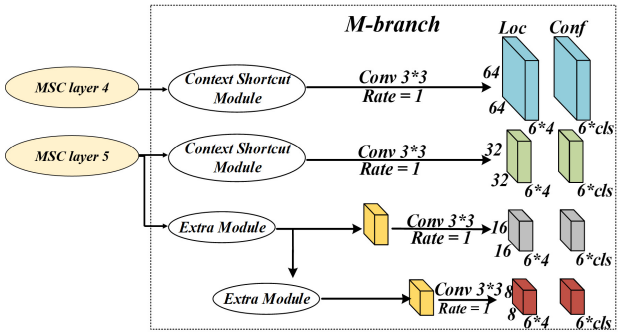


Fig. 8. Detailed structure of the medium-scale object regression branch layers in SIRLs.

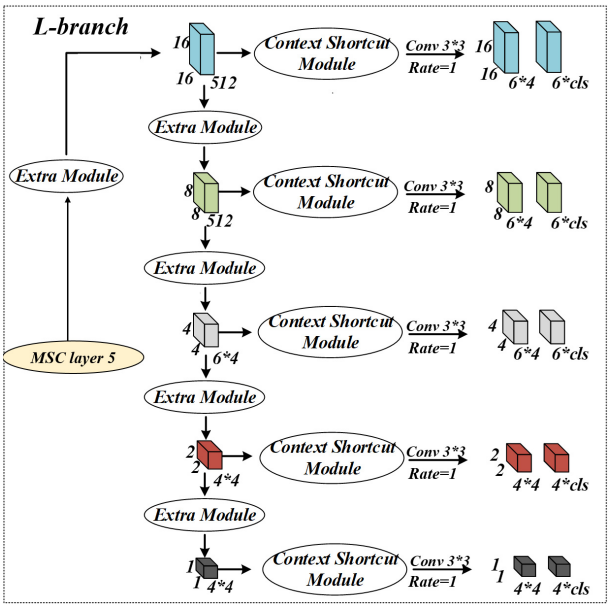


Fig. 9. Detailed structure of large-scale object regression branch layers in SIRLs.

than small objects considered in the S-branch. Therefore, the structure of the M-branch is different from the S-branch, as shown in the left side of Figs. 6(b) and 8. In Fig. 6(b), diversely colored lines represent the corresponding fine-scale ranges of the default bboxes, and pink rectangles represent the overlapping of scale compensation strategy in the M-branch. As presented in Fig. 8, MSC layers of 4 and 5 are individually fed into context shortcut modules, and a 3×3 convolution with Rate $= 1$ is used to generate the first and second regression layers. In addition, an extra module and a 3×3 convolution are applied to the output of MSC layer 5 to produce the third regression layer, and the same operations are then repeated to produce the fourth regression layer. Finally, these four generated regression layers are used for medium-scale object localization and classification regression.

L-branch: The L-branch is designed to detect objects with sqrt areas greater than 400 pixels. They are considered to be large objects in optical remote sensing images. To account for dramatic variations in object size in large scale, five regression layers with big size difference are designed for large object detection as shown in the left of Figs. 6(c) and 9. In the right of Fig. 6(c), diversely colored lines represent the corresponding fine-scale ranges of the default bboxes and the pink rectangles represent the overlapping of the scale compensation strategy in the L-branch. The details of the L-branch architecture are shown in Fig. 9, and these five regression layers are from MSC layer 5. First, MSC layer 5 is fed into an extra module. Then, the context shortcut module and 3×3 convolution are used to generate the first regression layer. Next, the remaining

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regression layers are generated by repeating these operations. Finally, these five regression layers can be used for large-scale object localization and classification regression. Moreover, the main reason for limited performances of large objects' detection is the fixed size of network input. Sometimes objects would be larger than the network input size, and these cases would make the IoU selection mismatch or get a fragmentary object information which would result in missed detection, poor object locating, or abundant false alarms. To address this problem, the multiscale training strategy is adopted, and the input images are down-sampled before being fed into the L-branch in both training and testing processes.

D. Specific Scale Joint Loss Function

Related to the structure of the SIRLs, the specific scale joint loss is proposed to make FSoD-Net comfortably converge. Here, the loss function used for multitask classification in each regression layer is the classical softmax, as shown in (4)

$$L_{\text{cls}}(x, c) = -\sum_{i\in\text{Pos}}^{N} x_{i,j}^{p}\log\left(\hat{c}_i^{p}\right) - \sum_{i\in\text{neg}}\log\left(\hat{c}_i^{0}\right) \tag{4}$$

where

$$\hat{c}_i^{p} = \frac{\exp\left(c_i^{p}\right)}{\sum_p \exp\left(c_i^{p}\right)}. \tag{5}$$

In (4), $x_{i,j}^{p} = \{0, 1\}$ is an indicator for the IoU threshold selected of the ith default bbox to the jth ground truth (GT) bbox of category p. If the default bbox is matched, the indicator is equal to "1"; otherwise, the indicator is equal to "0." In the default discrete scale bboxes' matching process, $\sum_i x_{i,j}^{p} \geq 1$ means that one or more default bboxes can be matched with GT bboxes, and N represents the total number of matched default bboxes. In (4) and (5), $\hat{c}_i^{p}$ denotes the corresponding predicted cth category score for the ith positive sample from the GT bboxes of category p, and $\hat{c}_i^{0}$ represents the background of class 0. Here, the number of unmatched bboxes would be much greater than the number of matched bboxes, resulting in a significant imbalance between the positive and negative training samples. Therefore, instead of using all negative examples, negative and positive samples are used at a ratio of 3:1 to achieve faster optimization and more stable training. Notably, softmax-based multitask classification produces a probability distribution (PD) consisting of a $\hat{c}_i^{p}$ value for each of the expected p object category. Here, $\sum_p \hat{c}_j^{p} = 1$, and the PDs can roughly present as two cases which are confusing and ideal cases.

In practical situations, the confusing cases occur frequently, which can lead to missed detections or false alarms. If the correct object category does not clearly have the highest confidence score, there is a high risk that the object would be detected as belonging to one of other categories or identified as background. Therefore, we consider to improve the detected confidence scores by constraining PD to be sparse. Then, an L_1-norm constraint is added to L_{cls} as a regularization term to force the PD generated by the softmax operation to be close to the ideal case, and the resulting expression is formulated as

$$L_{\text{cls}}^{1}(x, c) = -\sum_{i\in\text{Pos}}^{N}\left[x_{i,j}^{p}\log\left(\hat{c}_i^{p}\right) - \left\|\varphi\left(\hat{c}_i^{p}\right)\right\|_1\right] - \sum_{i\in\text{neg}}\left[\log\left(\hat{c}_i^{0}\right) - \left\|\varphi\left(\hat{c}_i^{0}\right)\right\|_1\right] \tag{6}$$

where

$$\varphi(c) = \text{PD}\left(\hat{c}_i^{p}\right) - \hat{y}^{p}. \tag{7}$$

In (7), $\text{PD}(\hat{c}_i^{p})$ represents the PD of $\hat{c}_i^{p}$. $\varphi(c)$ represents the comparison between the PD for the matched bboxes and the expected ideal $\hat{y}^{p}$. If the matched bboxes belong to the pth category, then the expected ideal PD is enforced to close to $\hat{y}^{p}$ by (7) and consists only of a confidence score of "1" for the pth category. Thus, by (6) and (7), the L_1-norm constraint can be used in multitask classification loss to not only improve the classification confidence scores but also avoid many false alarms.

In addition to multitask classification loss, the localization loss for the matched bboxes is also very important. Before regression based on the localization loss, coordinates' mapping is applied to prevent the effect of different object scales from impacting the results. The coordinates' mapping process is expressed as

$$\begin{aligned}
\hat{g}_j^{cx} &= \left(g_j^{cx} - b_i^{cx}\right)/b_i^{w}, \quad \hat{g}_j^{cy} = \left(g_j^{cy} - b_i^{cy}\right)/b_i^{h} \\
\hat{g}_j^{w} &= \log\left(\frac{g_j^{w}}{b_i^{w}}\right), \quad \hat{g}_j^{h} = \log\left(\frac{g_j^{h}}{b_i^{h}}\right) \\
\hat{l}_i^{cx} &= \left(l_i^{cx} - b_i^{cx}\right)/b_i^{w}, \quad \hat{l}_i^{cy} = \left(l_i^{cy} - b_i^{cy}\right)/b_i^{h} \\
\hat{l}_i^{w} &= \log\left(\frac{l_i^{w}}{b_i^{w}}\right), \quad \hat{l}_i^{h} = \log\left(\frac{l_i^{h}}{b_i^{h}}\right).
\end{aligned} \tag{8}$$

In (8), $m \in (cx, cy, w, h)$, where, cx, cy, w, and h denote the two center coordinates of a bbox and its width and height values, respectively. $\hat{l}_i^{m}$ is a vector representing four parameterized coordinates of the predicted bbox, and $\hat{g}_j^{m}$ is the corresponding GT vector. b_i^{m} is the parameterized coordinate vector of the default bbox. Then, the localization loss can be written as presented in (9)

$$L_{\text{loc}}(x, l, g) = \sum_{i\in\text{Pos}}^{N} x_{i,j}^{k}\left[R\left(\hat{l}_i^{cx} - \hat{g}_j^{cx}\right) + R\left(\hat{l}_i^{cy} - \hat{g}_j^{cy}\right)\right] + \sum_{i\in\text{Pos}}^{N} x_{i,j}^{k}\left[R\left(\hat{l}_i^{w} - \hat{g}_j^{w}\right) + R\left(\hat{l}_i^{h} - \hat{g}_j^{h}\right)\right] \tag{9}$$

where $x_{i,j}^{k}$ is an indicator that is defined in the same way as multitask classification loss designing, and $R(\cdot)$ is a robust loss function (smooth L_1 loss) defined in [5], which is formulated

$$R(x) = \begin{cases} 0.5x^2, & \text{if } |\text{x}| < 1 \\ |x| - 0.5, & \text{otherwise.} \end{cases} \tag{10}$$

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Then, based on the designed SIRLs' structure, the scale-specific joint loss function of FSoD-Net can be expressed

$$\begin{aligned} L &= L_{s(1)} + L_{s(2)} + L_{s(3)} \\ &= \frac{1}{N} \cdot \sum_{i=\mathrm{I}}^{\mathrm{III}} \left[L_{\mathrm{loc}}(x, l, g) + L_{\mathrm{cls}}^{1}(x, c) \right] \\ &= \frac{1}{N} \cdot \sum_{i=\mathrm{I}}^{\mathrm{III}} \left\{ \begin{array}{l} \sum\limits_{i \in \mathrm{Pos}}^{N} x_{i,j}^{k} R\left(\hat{l}_i^m - \hat{g}_j^m \right) \\ + \sum\limits_{i \in \mathrm{Pos}}^{N} \left[-x_{i,j}^{p} \log\left(\hat{c}_i^p \right) + \left\| \varphi\left(\hat{c}_i^p \right) \right\|_1 \right] \\ + \sum\limits_{i \in \mathrm{neg}} \left[-\log\left(\hat{c}_i^0 \right) + \left\| \varphi\left(\hat{c}_i^0 \right) \right\|_1 \right] \end{array} \right\} \end{aligned} \tag{11}$$

where S is an index representing the layers corresponding to different scales. Consistent with the SIRLs, the specific scale joint loss has three components, and each component includes localization loss and a special L_1-norm constrained multitask classification loss. Using (11), we can train the model to generate higher confidence scores and fewer false alarms in full-scale object detection followed by Algorithm 1.

Algorithm 1 One Stage End-to-End Training Procedure Using Stochastic Gradient Descent Optimization Method

Initialization Initialize branch weights W. Set learning rate $\mu > 0$. Training epoch $\mathrm{T} > 0$. Batch-size $\Theta > 0$. Backbone weights are pre-trained on NWPU-45 and remain unchanged.
Input: Training image I and label $\mathrm{t} = \{\mathrm{x_{min}}, \mathrm{y_{min}}, \mathrm{x_{max}}, \mathrm{y_{max}}\}$
Output: Branch weights W.
for $\mathrm{t} = 1, 2, 3, \ldots, \mathrm{T}$ **do**
  Randomly choose Θ samples as mini-batch;
  for $\theta = 1, 2, 3, \ldots, \Theta$ **do**
    Split $\mathbf{t}$ into three branch-isolated GT sets $\mathbf{t_s}$, $\mathbf{t_m}$, $\mathbf{t_l}$
    Generate $\mathrm{F_s}$, $\mathrm{F_m}$, $\mathrm{F_l}$; feature points from MSC layers
    Generate $\mathrm{A_s}/\mathrm{A_m}/\mathrm{A_l}$ default boxes in three isolated branches
    for $\mathrm{a} = 1, 2, \ldots, \mathrm{A_s}/\mathrm{A_m}/\mathrm{A_l}$ **do**
      Compute score and bounding box
      Compute Smooth L1 loss, Softmax loss and Specific Joint loss
    Compute $L = L_{s1} + L_{s2} + L_{s3}$
  Update branch weight W.
end

IV. Experiments and Analysis

A. Working Environment and Data Sets

All experiments were performed on an NVIDIA TITAN RTX GPU workstation with computer unified device architecture 10.0 (CUDA 10.0) and CUDA Deep Neural Network library 7.0 (cuDNN 7.0). The models were implemented in PyTorch 1.1.0 with Python 3.6. For evaluation, we used the standard mAP and mF1 scores calculated with an IoU threshold of 0.5. During the training phase, the input size of our model was 512 × 512 and the mini-batch size was set to 16. The learning rate was initialized as 0.004 and decreased by half in every 50 epochs. Except for testing the performance of FSoD-Net object detection, the classification ability of the lightweight MSE-Net backbone is also evaluated by *Kappa*, *Top1* accuracy, *Top3* accuracy, *Top5* accuracy, and confusion matrix. In addition, related to small-, medium-, and large-scale segments generated by K-means, each scale training data are up- or down-sampled into one scale to alleviate sample imbalance phenomenon in each fine-scale segment. Finally, these two challenging optical remote sensing object detection data sets of DOTA [47] and DIOR [48] and one optical remote sensing scene classification data set of Northwestern Polytechnical University-REmote Sensing Image Scene Classification 45 (NWPU-RESISC45) [70] are used to analyze the performances of comparison methods, and these data sets are introduced as below.

1) DOTA Data Set: DOTA [47] is a multiscale optical remote sensing data set containing images from various sensors and platforms, and each image has the dimensions of approximately 800 × 800 to 4000 × 4000 pixels. Then, DOTA [47] data set contains 188 282 object instances exhibiting a wide variety of scales, orientations, and shapes in 15 categories of optical remote sensing object (e.g., baseball diamond (BD), basketball court (BC), bridge (B), ground track field (GTF), harbor (HB), helicopter (H), large vehicle (LV), plane (P), roundabout (R), ship (S), small vehicle (SV), soccer ball field (SBF), storage tank (ST), swimming pool (SP), and tennis court (TC)). Images included are from the Google Earth service and GF-2 and JL-1 satellites. Next, the training set of DOTA [47] data set contains 1411 remote sensing images, and the validation set contains 458 images with various object scale ranges.

2) DIOR Data Set: DIOR [48] is another optical remote sensing object detection data set, and it has the largest scale on both numbers of images and object categories. Then, DIOR [48] data set contains 23 463 optical remote sensing images and 192 472 object instances covering 20 common object categories [e.g., airplane (AL), airport (AT), baseball field (BF), BC, bridge (B), chimney (C), dam (D), expressway service area (ESA), expressway toll station (ETS), golf course (GC), GTF, HB, overpass (O), ship (S), stadium (SD), ST, TC, train station (TS), vehicle (V), and wind mill (WM)]. Then, the size of images which are collected from Google Earth service in DIOR [48] is 800 × 800 pixels and the spatial resolution ranges from 0.5 to 30 m. Furthermore, the images in DIOR [48] have different imaging conditions, weathers, seasons, and qualities. These 20 classes' instances own high interclass similarity, intraclass diversity, and size variances, not only in terms of spatial resolution but also in the aspect of inter- and intraclass size variability across objects.

3) NWPU-RESISC45: NWPU-RESISC45 [70] is currently the largest data set for scene classification which consists of 45 scene categories with 700 images per class. Then, it comprises 31 500 scene images with 256 × 256 pixels, and the spatial resolution varies from about 0.2 to 30 m. Therefore, NWPU-RESISC45 [70] is very suitable to be used to explore the classification ability of different backbones which are usually used for one- or two-stage object detectors.

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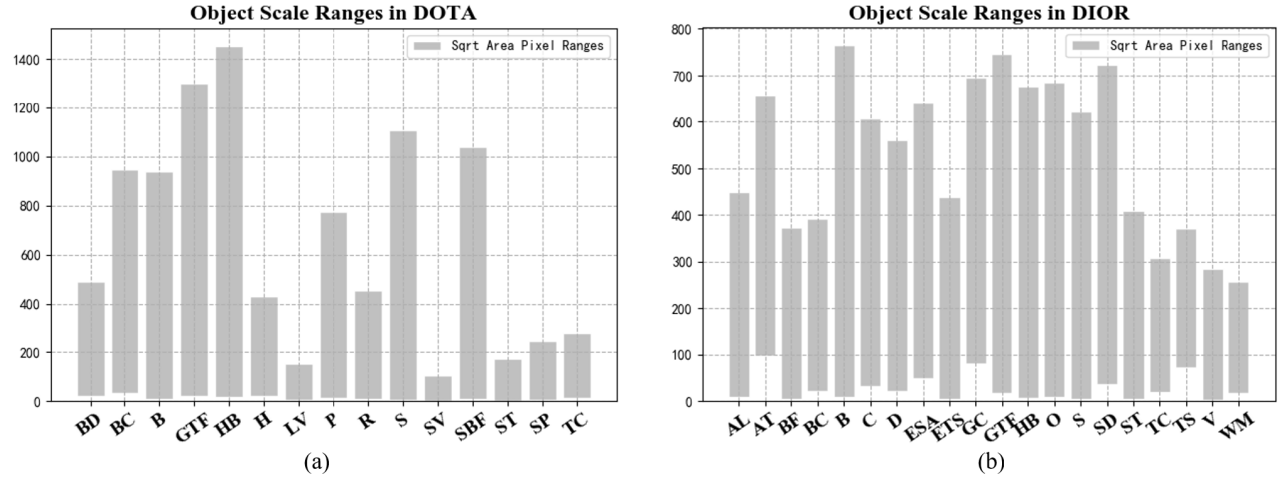


Fig. 10. Object scale range distributions in (a) DOTA [47] and (b) DIOR [48] data sets.

TABLE II
PERFORMANCE AP (%) AND MAP (%) OF DIFFERENT METHODS ON DOTA DATA SET

Method	mAP	mF1	*BD*	*BC*	*B*	*GTF*	*HB*	*H*	*LV*	*P*	*R*	*S*	*SV*	*SBF*	*ST*	*SP*	*TC*
YOLT [64]	61.44	**0.65**	60.50	70.93	22.09	37.54	56.58	22.16	70.48	87.20	57.62	84.43	70.75	52.70	78.96	55.76	93.94
Faster-RCNN [20]	56.13	0.47	63.70	50.10	29.60	54.86	65.71	48.37	62.59	76.15	40.91	67.70	28.59	55.84	67.33	43.64	86.89
HSF-Net [34]	62.36	0.62	69.88	69.40	37.70	57.98	66.84	41.78	64.24	79.96	52.81	71.84	66.76	47.47	61.79	59.15	87.87
SSD [16]	50.19	0.43	32.83	39.40	23.02	38.82	65.21	22.16	62.84	85.37	34.83	52.31	33.44	52.29	67.33	55.73	87.30
Yolo-v3 [28]	65.67	0.52	68.78	66.82	45.93	51.92	74.03	56.68	60.67	93.91	45.01	85.56	50.12	52.45	83.47	55.85	93.88
Yolo-v4 [29]	67.97	0.60	72.82	60.04	70.00	72.27	64.53	60.21	67.39	77.09	79.29	78.97	49.76	61.45	60.96	60.97	83.81
RFB-Net [19]	64.03	0.49	49.76	63.83	35.74	36.35	63.21	20.53	77.53	87.75	67.93	90.05	78.00	33.57	89.54	72.78	93.84
RetinaNet [18]	63.27	0.58	68.17	62.33	65.96	**76.22**	65.96	48.50	62.78	72.99	**80.90**	68.59	25.51	57.78	51.31	57.81	84.20
CenterNet [25]	73.94	0.63	**78.56**	66.11	45.39	53.39	**78.86**	**66.82**	80.24	**97.37**	69.02	90.30	62.16	**64.86**	85.75	75.63	**94.58**
EfficientDet D0 [30]	61.96	0.57	64.63	64.63	44.22	72.25	51.14	55.05	52.39	93.09	42.74	81.57	38.87	52.12	78.27	44.54	92.18
EfficientDet D2 [30]	68.61	0.60	73.60	60.51	72.14	74.62	64.57	60.50	67.38	77.37	80.28	78.64	50.46	61.82	61.59	62.11	83.58
EfficientDet D4 [30]	70.72	0.62	72.23	64.01	**73.26**	73.23	65.35	62.02	63.15	93.27	79.94	85.61	59.70	40.92	75.94	59.56	92.66
S3FD [26]	30.22	0.31	30.60	36.43	23.71	25.12	33.91	9.11	26.92	36.33	16.51	27.01	25.11	28.52	27.20	26.23	81.01
M2Det [27]	34.48	0.43	59.34	43.91	27.42	40.63	35.22	18.21	26.91	36.13	49.84	18.11	9.11	35.32	27.33	17.31	72.70
FSoD-Net (Our)	**75.33**	**0.65**	71.13	**78.93**	51.34	57.46	71.34	64.86	**82.15**	93.11	70.64	**92.39**	**78.18**	53.32	**93.48**	**78.69**	92.87

Furthermore to see each instance scale variance in DOTA [47] and DIOR [48] data sets, the object scale range distributions in DOTA [47] and DIOR [48] are presented in Fig. 10. From Fig. 10, we can see that DOTA [47] and DIOR [48] have diversity or vast object scale variances, and thus, they are very suitable for demonstrating the significant improvements achieved with the proposed full-scale object detection method.

B. Evaluation Indexes

To evaluate the proposed FSoD-Net, mAP and mF1 [16]–[30] scores were calculated to assess the performance of multiscale object detection in optical remote sensing images. In addition, AP and precision–recall (PR) curves were used to evaluate each class object detection performance. Therefore, before the performance discussion, several evaluation indexes must be explained in this section. The IoU is the overlap ratio between the predicted bboxes and the corresponding GTs, and it is an important parameter in anchor-based object detection methods. During the training procedure, the trained bboxes are filtered based on the IoU which is defined as follows:

$$\text{IoU} = \left(S_{\text{bbox}} \cap S_{\text{gt}}\right) / \left(S_{\text{bbox}} \cup S_{\text{gt}}\right) \tag{12}$$

where S denotes the pixel areas of bbox. The value of true positives (TPs) is the number of correctly detected targets, for which the IoU between the predicted S_{bbox} and corresponding S_{gt} is higher than 0.5 and the class of the target is correctly predicted. Similarly, the value of false positives (FPs) is the number of incorrectly predicted objects, for which the IoU between the predicted S_{bbox} and S_{gt} is higher than 0.5, but the object class is predicted incorrect. The value of false negatives (FNs) is the number of targets whose detection is missed, meaning that an actually existing object is not matched with any predicted bbox. On the basis of these metrics, the precision and recall can be formulated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{13}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{14}$$

Furthermore, to evaluate the classification ability of the proposed lightweight MSE-Net, the indexes of *Kappa coefficient*, *Top* 1 accuracy, *Top* 3 accuracy, *Top* 5 accuracy, and confusion matrix [49]–[57] are also used.

C. Comparative Analysis of Results

To significantly indicate prominent performance and contribution of the proposed FSoD-Net, several famous one-stage CNN-based detectors (e.g., SSD [16], RetinaNet [18], RFB-Net [19], Yolo-v3 [28], YOLT [64], Yolo-v4 [29], CenterNet [25], single shot scale-invariant face detector (S3FD) [26],

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TABLE III

PERFORMANCE AP (%) AND MAP (%) OF DIFFERENT METHODS ON DIOR DATA SET

Method	mAP	mF1	*AL*	*AT*	*BF*	*BC*	*B*	*C*	*D*	*ESA*	*ETS*	*GC*	*GTF*	*HB*	*O*	*S*	*SD*	*ST*	*TC*	*TS*	*V*	*WM*
YOLT [64]	60.3	0.64	64.7	69.0	32.8	87.8	32.3	71.5	45.8	54.9	55.8	65.6	66.3	49.9	49.9	87.7	30.3	73.3	82.0	29.9	52.4	74.0
Faster-RCNN [20]	53.6	0.36	51.4	61.6	62.2	80.7	27.0	74.2	37.3	53.4	45.1	69.6	61.8	43.7	49.0	56.1	41.8	39.6	73.9	44.7	34.0	65.3
HSF-Net [34]	54.9	0.64	51.0	62.1	60.2	80.7	27.4	72.8	43.8	52.1	45.2	67.5	58.7	34.0	48.9	70.8	45.5	53.5	72.9	46.8	37.4	67.0
SSD [16]	60.9	0.45	92.9	48.3	77.9	77.2	35.7	66.6	46.5	47.9	58.5	37.3	53.7	47.9	39.9	59.6	77.5	**80.8**	88.3	55.6	38.8	86.7
Yolo-v3 [28]	59.9	0.62	67.5	54.7	65.8	86.8	34.2	73.5	34.3	55.7	49.6	67.3	68.9	**54.3**	51.7	86.8	40.3	67.8	83.9	32.3	49.1	73.6
Yolo-v4 [29]	65.9	0.69	78.3	**69.7**	69.6	88.3	38.7	76.9	48.6	57.6	57.9	**74.1**	70.6	52.8	56.8	**89.4**	50.4	77.5	83.3	46.0	**55.7**	75.9
RFB-Net [19]	62.6	0.57	**93.8**	48.3	80.2	77.7	40.7	67.5	47.4	48.5	70.0	37.3	54.9	48.2	43.1	66.1	78.0	80.0	89.1	**55.6**	40.4	88.3
RetinaNet [18]	65.2	0.50	63.4	47.8	83.3	78.4	48.2	67.9	49.4	53.2	73.6	66.3	59.1	47.6	45.7	72.0	82.4	80.7	90.1	55.0	47.7	92.9
CenterNet [25]	61.6	0.59	64.0	66.3	65.7	86.3	34.8	73.1	41.1	60.8	54.2	73.0	66.0	45.3	53.3	81.3	53.5	63.7	80.9	44.1	46.3	78.8
EfficientDet D0 [30]	59.9	0.46	91.9	47.3	79.2	76.0	36.0	66.5	46.8	65.6	47.5	37.0	53.3	45.8	39.8	54.9	77.2	76.3	87.3	54.4	31.9	85.1
EfficientDet D2 [30]	61.5	0.55	72.5	66.2	64.3	87.0	33.2	74.0	43.1	71.8	54.3	55.6	52.7	37.4	47.0	86.1	66.8	70.9	81.1	50.2	43.3	73.0
EfficientDet D4 [30]	66.1	0.61	86.5	57.4	75.7	85.2	33.5	75.4	**65.6**	80.1	67.4	58.3	71.4	35.6	50.6	78.8	**90.3**	61.8	82.9	54.6	30.0	81.5
S3FD [26]	54.7	0.58	71.7	46.3	72.7	75.5	31.0	71.9	47.9	51.7	49.6	67.2	70.9	23.0	45.1	44.3	61.3	43.9	80.0	30.0	34.3	74.8
M2Det [27]	57.6	0.53	63.3	67.6	71.1	80.2	32.1	72.5	58.0	62.3	52.5	69.8	63.5	41.3	52.0	35.9	66.3	35.9	72.2	52.4	25.0	78.3
FSoD-Net(Our)	**71.8**	**0.70**	88.9	66.9	**86.8**	**90.2**	45.5	**79.6**	48.2	**86.9**	**75.5**	67.0	**77.3**	53.6	**59.7**	78.3	69.9	75.0	**91.4**	52.3	52.0	**90.6**

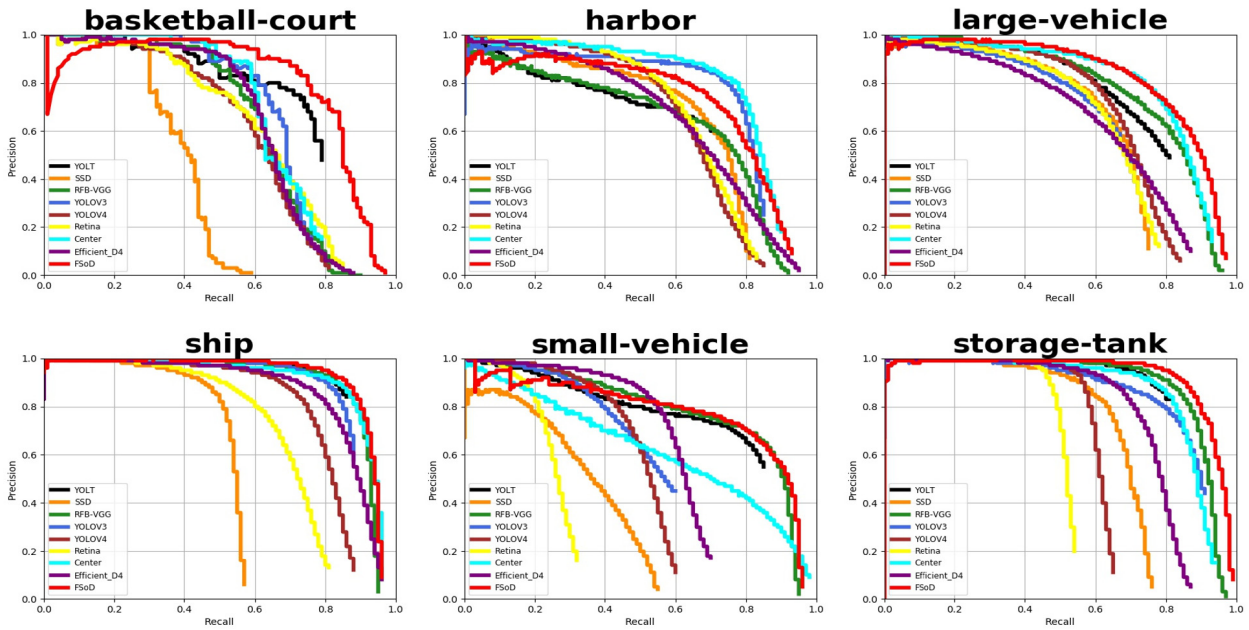


Fig. 11. $P-R$ curves analysis on the DOTA [47] data set.

M2Det [27], and three versions D0, D2, and D4 of Effcient-Det [30]) are used as comparisons. In addition, some two-stage detectors (e.g., Faster-region-CNN (RCNN) [20] and HSF-Net [34]) are also selected. All comparisons are tested on DOTA [47] and DIOR [48] data sets, and the results are reported in Tables II and III. Then, for further revealing full-scale object detection performance of the proposed FSoD-Net, there are some $P-R$ curves shown in Figs. 11 and 12, and meanwhile, some visualization detection results of FSoD-Net are shown in Figs. 13 and 14. Except these, we also evaluate the inference speed and Flops of comparisons in Table IV to show that proposed the FSoD-Net is a more accurate and effective one-stage detector applied in optical remote sensing object detection than the state-of-the-art methods.

1) Results on DOTA Data Set: From Fig. 10, we can see that the DOTA [47] data set contains 15 category instances with a great scale variance than the DIOR [48] data set, and thus it is very suitable for demonstrating the full-scale object detection performance from optical remote sensing images. In Table II, there are the reported comparison results on DOTA [47], and we can see that the proposed FSoD-Net performs better detection indexes of 75.33% mAP and 65% mF1 than other state-of-the-art methods, whether one-stage or classical two-stage detectors. First, in the second line of Table II, the classical two-stage detector of Fast-RCNN [20] provided a bad performance on DOTA [47] data set, because it used the backbone last feature layer for region proposal which is hard to provide the accurate localization information of tiny objects in deep feature layer (e.g., 28.59% AP for SV detection). Then, in the third line of Table II, HSF-Net [34] used several different sizes of convolutions applied for the last feature layer of the backbone to construct the multiscale layers like FPN for region proposal in Faster-RCNN [20] framework. As the result reported, although HSF-Net [34] has certain improvement compared with Faster-RCNN [20], it cannot deal with the large-scale span object detection (e.g., S, HB, and BC shown in Fig. 10). Thus, HSF-Net [34] and Faster-RCNN [20] all provided bad performances than the proposed FSoD-Net on the DOTA [47] data set. Second, from Table II, several famous one-stage detectors such as SSD [16], RetinaNet [18], RFB-Net [19], Yolo-v3 [28], YOLT [64], and Yolo-v4 [29] also provide unsatisfied detection results, because their regression and prediction layers like FPN which is the same as HSF-Net [34] cannot provide abundant receipt field

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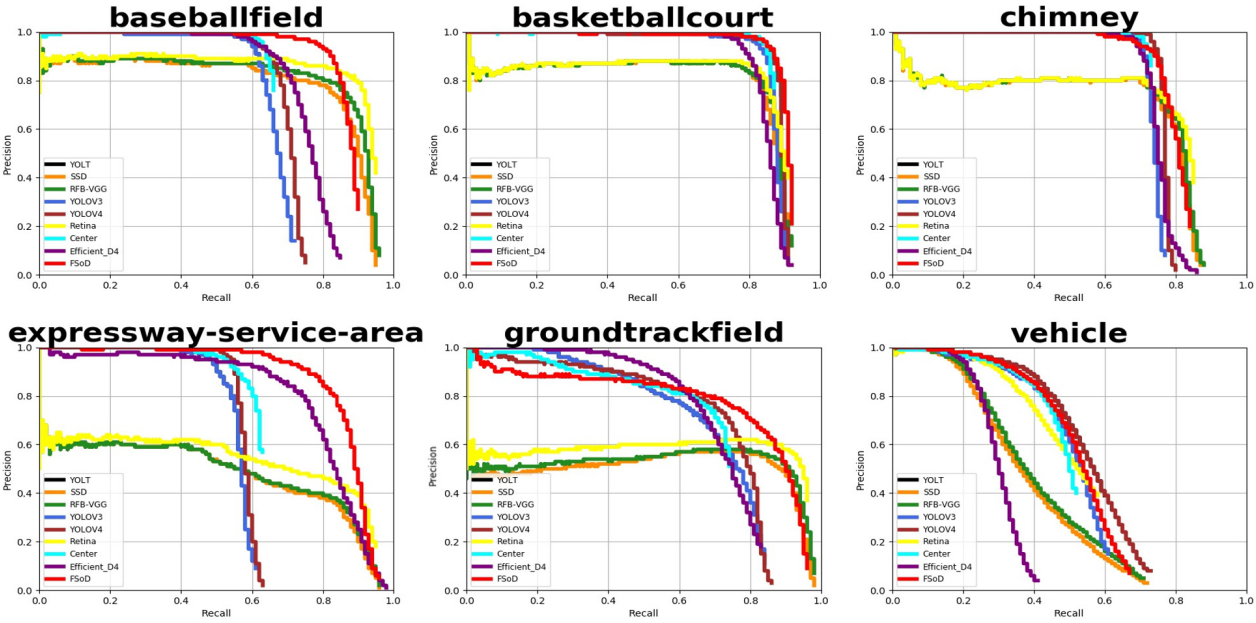


Fig. 12. $P-R$ curves analysis on the DIOR [48] data set.

supported multiscale object description within the large-scale span. Thus, related to specific FPN-like structure, the predefined bboxes of general one-stage detectors have a big risk to occur under- or over-fitting phenomenon of continuous object scale description within the large-scale span (e.g., some scale ranges are ignored or insufficient scale samples are selected for model training which is discussed in Section III-A). Third, as shown in Table II from lines 10 to 12, three versions D0, D2, and D4 of EfficientDet [30] provide 61.96%, 68.61%, and 70.72% mAP on DOTA [47]. Although the BiFPN structure in EfficientDet [30] can leverage the multiscale feature propagation and fusion, the complex and redundant feature fusion procedure would lead unpredictable scale feature interferences and make EfficientDet [30] difficult to converge.

In addition, in Table II line 9, one-stage anchor-free-based detector CenterNet [25] is also tested on DOTA [47], which can obtain 73.94% mAP and 63% mF1. Although this anchor-free-based CenterNet [25] can prevent complex bboxes' initialization and regression process, how to achieve accurate pixel-level center points' prediction and generate corresponding multiscale detection bboxes directly constraint its detection performance. Furthermore, the recently proposed nature scene multiscale one-stage detectors of M2Det [27] and S3FD [26] are chosen as the comparisons. Due to complex background interferences and continuous object scale description within the large-scale span challenges in optical remote sensing scenes, S3FD [26] and M2Det [27] cannot exhibit good performances. Finally, to further exhibit the proposed FSoD-Net excellent performance, there are several instances of $P-R$ curves and detection results shown in Figs. 11 and 13. From these $P-R$ curves in Fig. 11, we can see that these red curves of FSoD-Net can provide better full-scale object (e.g., ship, HB and basketball count) detection results than the state-of-the-art CNN-based one-stage detectors, especially for tiny object detection (e.g., SV, ST, and LV). Next, from Fig. 13, we can see that FSoD-Net can provide more accurate detection results with less false alarms in global and local optical remote sensing scenes.

2) Results on DIOR Data Set: As illustrated in Fig. 10, the DIOR [48] data set includes more category instances than the DOTA [47] data set, and it makes the challenge for classification task in object detection. Therefore, the DIOR [48] data set is also used for testing FSoD-Net and comparison methods. Then, from the overall results in Table III, FSoD-Net can produce the highest 71.8% mAP and 0.70 mF1 on DIOR [48], and these results indicate that the proposed lightweight MSE-Net backbone in FSoD-Net has a powerful classification ability than VGG-16 [52], ResNet-101 [53], and EfficientNet [57] dealing with 20 category instances and unseen background interferences from optical remote sensing scenes. Next, we also selected vast scale changed and tiny objects (i.e., BF, BC, chimney, express way–service area GTF, and vehicle) in DIOR [48] to prove the superior performance of FSoD-Net, and as shown in Fig. 12, these red $P-R$ curves indicate that the proposed FSoD-Net can also obtain better performances with selected objects. Then, there are several visualized results of the proposed FSoD-Net on DIOR [48] shown in Fig. 14, and we can see that whether for large-, medium-, or small-scale object, FSoD-Net also can provide excellent detection results.

3) Inference Speed Analysis: In this part, the inference speed and Flops of the proposed FSoD-Net and comparisons are tested and shown in Table IV. From Table IV, the FSoD-Net can get the acceptable inference time of 55 ms per image with the best detection performance on the DOTA [47] and DIOR [48] data sets. Then, since we use a lot of parallel convolutions and cascaded the small size convolution kernels instead of a large convolution kernel in the MSE-Net backbone, and then, due to the fine-scale ranges' regression and prediction of SIRLs' structure, the proposed FSoD-Net has 232.2M parameters which are less than Yolo-v3 [28] and Yolo-v4 [29] with the 1024×1024 input. In general, as results

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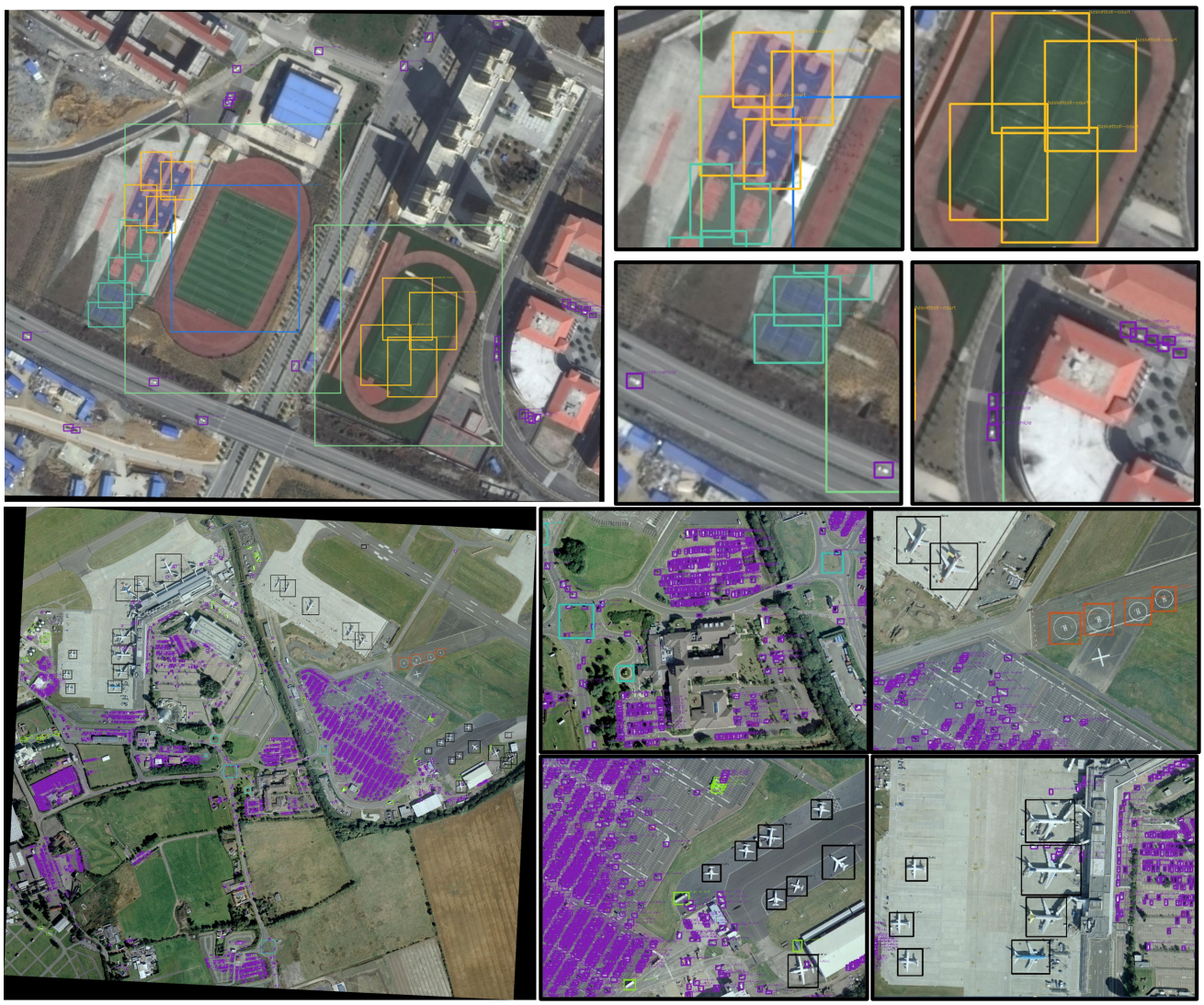


Fig. 13. Global and local detection results of the proposed FSoD-Net on DOTA [47]. The orange, blue-green, blue, and green bboxes represent the BC, TC, SBF, and GTF correspondingly. The dark green and light green bboxes represent the SV and LV, respectively. The black and blue bboxes represent the planes and roundabouts, respectively. All confidence scores are higher than 0.1.

TABLE IV
INFERENCE TIME AND FLOPS ANALYSIS OF DIFFERENT METHODS

Method	Backbone	Inference Time	DOTA@mAP	DIOR@mAP	FLOPs	Model size(M)
FSoD Net 512 Ours	MSE-Net	**55 ms**	**75.33**	**71.80**	165B	232.2
RFBNet 512 [19]	VGG16	30 ms	64.03	62.59	102B	185.1
SSD 512 [16]	VGG16	28 ms	50.19	60.87	82B	146.7
Yolov3 1024 [28]	Darknet53	29 ms	65.67	59.93	71B	240.0
Yolov4 1024 [29]	CSPD Darknet53	26 ms	67.97	65.91	221B	244.6
RetinaNet 800 [18]	Resnet101	90 ms	63.27	65.22	248B	147.2
EfficientDet D4 [30]	EffecientNet	42 ms	70.72	66.13	55B	83.9

in Table IV, the proposed FSoD-Net not only produces the best performance of object detection in the DOTA [47] and DIOR [48] data sets but also has acceptable parameters and inference speed.

D. Optimal Parameter Discussion

1) Backbone Analysis: To demonstrate the effectiveness of the proposed backbone MSE-Net, several famous backbone networks which are usually used for one- or two-stage detectors including VGG-16 [52], ResNeXt-29 [55], ResNet101 [53], ResNeXt-101 [55], DetNet [54], ResNesSt [56], and EfficientNet versions from b0 to b4 [57] are used as comparisons and carried out on the NWPU-RESISC45 [70] data set. As shown in Table V lines 6 and 7, ResNeXt-101 [55] just obtained 3.09% and 3.2% improvements of *Top* 1 accuracy and *Kappa coefficient* than ResNeXt-50 [55], which indicates that increasing the depth of backbone cannot bring a lot of performance improvements. Besides, the width, depth, and resolution are the three main elements to affect the backbone classification performance.

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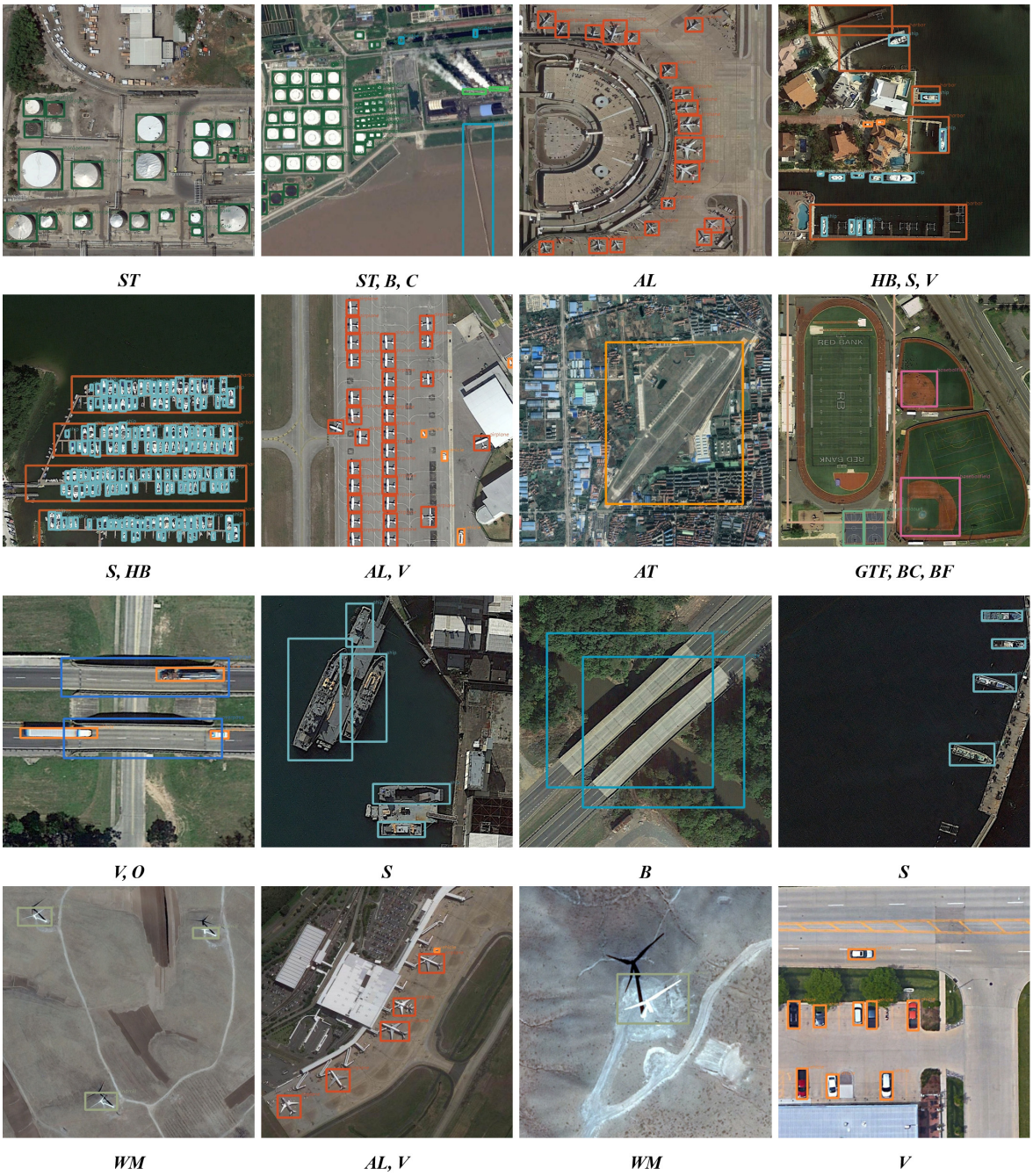


Fig. 14. Visualization detection results of FSoD-Net on the DIOR [48] test set. All confidence scores are higher than 0.1.

Thus, we also select DetNet [54] and EfficientNet versions b0–b4 [57] as comparisons. Here, DetNet [54] is focused on keeping higher spatial resolution in deep layers to improve its performance, and then the EfficientNet versions b0–b4 [57] comprehensively considered these depth, width, and resolution by scaling factors to enhance their performance. However, whether DetNet [54] or EfficientNet versions b0–b4 [57], they all provide lower *Top 1*, *Top 3*, *Top* 5 accuracy, and *kappa coefficient* than the proposed lightweight MSE-Net. In addition, as reported in Table V line 8, ResNeSt-101 [56] also exhibits better classification performance, but it has many parameters with deep layers. Although the backbone network classification performance can be improved by optimizing the width, depth, and resolution of the network, suitable input enhancing is also important for feature extraction. Thus, from Table V lines 12 and 13, we can see that in the proposed lightweight MSE-Net, the trainable Laplace kernel is designed for enhancing weak or tiny object feature information reasonably by the SE module. Next, the proposed lightweight MSE-Net also pays more attention to the width of the network to improve the multiscale feature extraction ability in fewer convolution layers, which is friendly in building suitable regression layers to trade-off tiny, medium, and large object location and classification regression. In summary, the proposed lightweight MSE-Net backbone can provide a superior performance than the other state-of-the-art backbones

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TABLE V
CLASSIFICATION ABILITY AND MODEL SIZE OF DIFFERENT BACKBONE NETWORKS

Backbone Network	*Top1 Accuracy*	*Top3 Accuracy*	*Top5 Accuracy*	*Model size（MB)*	*Kappa Coefficient*
EfficientNet b0 [57]	0.858413	0.960741	0.979365	16.5	0.855
EfficientNet b1 [57]	0.863915	0.961799	0.980423	26.6	0.861
EfficientNet b2 [57]	0.864021	0.965503	0.982011	31.4	0.861
EfficientNet b3 [57]	0.869206	0.962434	0.980106	43.5	0.866
EfficientNet b4 [57]	0.875767	0.967407	0.982751	71.2	0.873
ResNeXt50 [55]	0.878857	0.966984	0.985016	92.7	0.876
ResNeXt101 [55]	0.909841	0.978307	0.990159	169.6	0.908
ResNeSt101 [56]	0.937778	0.989841	0.995026	186.1	0.936
ResNet101 [53]	0.845270	0.957651	0.978222	172.0	0.842
DetNet [54]	0.832275	0.945926	0.972063	69.9	0.828
VGG16 [52]	0.828571	0.946138	0.972698	126.1	0.825
MSE-Net (without SE module)	0.914921	0.984399	0.993333	34.2	0.913
MSE-Net (our)	**0.933112**	**0.991217**	**0.997566**	47.4	**0.932**

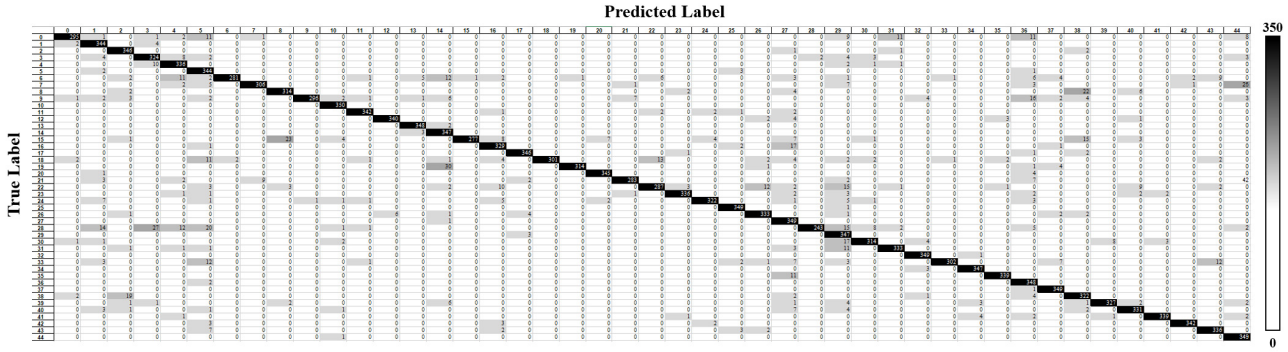


Fig. 15. Confusion matrix of our proposed lightweight MSE-Net on NWPU-RESISC45 [70] scene classification data set.

TABLE VI
ABLATION STUDY WITH DIFFERENT COMPONENTS SETTING ON DIOR DATA SET

Method	**Backbone**				**Regression layer**			**Scale Joint loss**	**mAP@DIOR(%)**
	VGG-16		MSE-Net		SIRLs				
	No SE	*SE*	*No SE*	*SE*	*Small*	*Medium*	*Large*		
Baseline	✓								62.60
		✓							64.07
			✓						64.58
				✓					65.17
				✓	✓				67.00
				✓	✓	✓			68.79
				✓	✓	✓	✓		71.06
FSoD-Net				✓	✓	✓	✓	✓	71.80

with fewer parameters. Moreover, for exhibiting classification performance of lightweight MSE-Net on NWPU-RESISC45 [70], the confusion matrix of MSE-Net is shown in Fig. 15.

2) Componentwise Analysis: To further demonstrate the effectiveness of the proposed FSoD-Net, the designed MSE-Net backbone and SIRLs with L_1-norm constraint scale joint loss components are tested on the DIOR data set [48], and the results are shown in Table VI. First, we select RFB-Net [19] as the baseline method, because RFB-Net [19] is a powerful one-stage detector which also refers to the SSD [16] framework (i.e., RFB-Net [19] used VGG-16 [52] backbone and multiscale feature layers like FPN [21] structure achieving regression and prediction) and has a good tiny object detection performance. From the first line of Table VI, we can see that based on the VGG-16 [52] backbone and FPN-like regression layers, the baseline method can provide 62.60% mAP on the DIOR [48] data set. Next, between the first and second lines or the third and fourth lines of Table VI, the proposed SE module is added into VGG-16 [52] or MSE-Net, and it can obtain 1.47% or 0.49% mAP improvements on DIOR [48]. Then, from the third and fourth lines, we can see that when the backbone of baseline is replaced by the proposed MSE-Net, it can reach 65.17% mAP than the basic 62.60% mAP. Next, from the fifth, sixth, and seventh lines in Table VI, the small-, medium-, and large-scale branches are gradually tested to exhibit the performance of the proposed SIRLs, and from these results of three lines, we can see that compared with the FPN-like structure in baseline, SIRLs can facilitate 5.89% mAP improvements and reach 71.06% mAP on the DIOR [48] data set. Consequently, suitable regression layers are important

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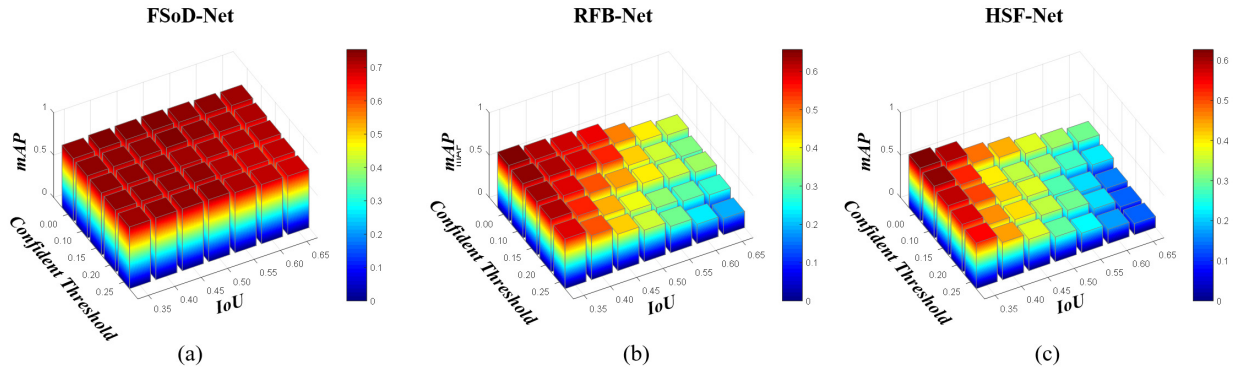


Fig. 16. IoU and confidence score threshold discussion of the proposed (a) FSoD-Net, (b) RFB-Net [19], and (c) HSF-Net [34].

in achieving a continuous scale description for object detection within a large-scale span. Finally, the L_1 constraint scale joint loss is discussed in the last line of Table VI, and compared with the results of the last two lines, the proposed scale joint loss can get 0.47% mAP improvements and further leverage the proposed FSoD-Net to obtain the state-of-the-art 71.80% mAP on the DIOR [48] data set.

3) IoU and Confident Value Setting Discussion: For either one- or two-stage CNN-based detectors, IoU is a very important parameter in training and testing procedures. In the training procedure, the strict constraint of the IoU threshold (i.e., IoU $=$ 0.7) can provide accurate object-level location and classification information; however, it would lead insufficient training samples to be selected for model training. On the other hand, although lower IoU threshold (i.e., IoU $=$ 0.3) can ensure sufficient training samples to be selected for model training, most of the selected training samples are invalid with incomplete object-level information which would severely affect location and classification regression process and make convergence more difficultly. Therefore, we used the DOTA [47] data set to discuss the effects of the IoU threshold setting for the proposed FSoD-Net and comparisons. As illustrated in Fig. 16(a), we can see that the performance of the proposed FSoD-Net is only slightly affected by modifying IoU threshold. On the contrary, the one-stage RFB-Net [19] and two-stage HSF-Net [34] show severe declines in mAP when IoU threshold becomes strict constraints as shown in Fig. 16(b) and (c). Then, in Fig. 16(b) and (c), without effective design of regression layers and scale compensation strategy of default bboxes, only a few default bboxes corresponding to GTs can be selected for model training relatively, and thus, they cannot provide good detection results. Owing to the proposed SIRLs' structure in combination with scale compensation strategy, FSoD-Net is not sensitive to the strict constraint condition of the IoU threshold, and when the IoU threshold is equal to 0.5, FSoD-Net performs the best performance. In addition, with the different output confidence thresholds, the proposed FSoD-Net object detection method also maintains a higher performance than RFB-Net [19] and HSF-Net [34], because the proposed L_1-norm constraint scale joint loss in each scale layer can urge the detected objects having the higher confident value toward "1."

V. Conclusion

In this article, a novel one-stage object detection network named FSoD-Net is proposed. First, the FSoD-Net includes a powerful lightweight MSE-Net backbone which can facilitate basic feature extraction for tiny and multiscale objects. Based on the powerful lightweight MSE-Net backbone, the SIRLs' structure with scale compensation strategy is proposed to make optical remote sensing object with large-scale span regress effectively. Then, related to SIRLs' structure, a specific scale joint loss with an L_1-norm constraint is proposed to speed up the convergence and improve the classification scores of predicted bboxes. Next, challenging DOTA [47] and DIOR [48] data sets for optical remote sensing object detection are used to demonstrate that the proposed FSoD-Net can achieve better detection performance than the other state-of-the-art one-stage detectors. Nevertheless, the extensive experiments show that higher detection ratios of the proposed FSoD-Net are accompanied by inevitable false alarm phenomenon. In addition, when the training samples are insufficient or unbalanced, FSoD-Net also shows a bad performance. Therefore, our future work will focus on the false alarm elimination and sample imbalance training problem to improve one-stage detector's performance in optical remote sensing full-scale object detection.

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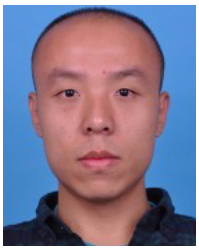
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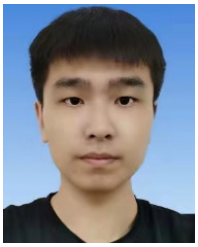
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